

Lucid Dream Control Questionnaire (LDCQ): Development and Validation

Emma Peters¹, Daniel Erlacher¹, Clarita Bonamino²

¹ Institute of Sports Science, University of Bern, Bremgartenstrasse 145, 3012 Bern, Switzerland

² School of Exercise and Nutrition Sciences, University of Technology Queensland, Kelvin Grove Campus, 44 Musk Avenue, Kelvin Grove, QLD 4059, Australia

Corresponding Author:

Emma Peters

emma.peters@unibe.ch

Emma Peters  <https://orcid.org/0000-0003-3612-0365>

Daniel Erlacher  <https://orcid.org/0000-0001-8981-2686>

Clarita Bonamino  <https://orcid.org/0000-0002-8843-4211>

Abstract

Lucid dream control (LDC), the ability to deliberately influence dream experiences during a lucid dream, is central to many theoretical, experimental, and lucid dream clinical applications. Despite its importance, standardized instruments for assessing LDC remain limited, typically treating it as a single, broad construct. This study aimed to develop and validate the Lucid Dream Control Questionnaire (LDCQ), a multidimensional measure assessing seven theoretically driven domains of volitional LDC: Own Cognition, Own Emotion, Own Body, Own Movement, Characters, Environment, and Physics. Psychometric evaluation demonstrated good internal consistency across all domains, meaningful item-total associations, and strong convergent validity with the dream control composite of the Lucid Dreaming Skills Questionnaire (LUSK) (an existing validated measure of lucid dreaming skills) ($r = .73, p < .001$). Exploratory construct validity further showed positive associations between LDCQ scores and lucid dream ($r = .63, p < .001$) and dream recall frequency ($r = .39, p < .001$), supporting preliminary construct validity. Quantitative response patterns and qualitative participant feedback further informed questionnaire refinement, resulting in a 36-item instrument. The final scoring framework introduces domain-specific dream control scores, dream engagement scores, and a multidimensional LDC profile, allowing characterization of both perceived control ability and breadth of dream control behaviors attempted. An accompanying automated online scoring and visualization tool was also developed to facilitate research and practical application. Overall, the LDCQ provides a theoretically grounded and psychometrically supported instrument for investigating individual differences in LDC across research and applied contexts.

Keywords: *lucid dreaming, lucid dream control, questionnaire validation, dream control profile, psychometrics, dream engagement*

23 Highlights

- 24 – Developed and validated LDCQ, a multidimensional lucid dream control questionnaire
- 25 – The LDCQ assesses seven distinct domains of volitional lucid dream control
- 26 – Strong reliability and convergent validity support LDCQ's psychometric quality
- 27 – Dream Engagement and Dream Control Profile extend traditional questionnaire scoring
- 28 – Online scoring tool enables automated profile generation and visualization

1. Introduction

A lucid dream (LD) is a dream where one becomes aware that they are dreaming while the dream continues (LaBerge, 1985). This state provides a unique opportunity to study consciousness, metacognition, and agency during sleep, as dreamers become capable of reflecting on the dream state and may intentionally control aspects of the ongoing dream. However, awareness of the dream state alone does not necessarily confer control over the dream. Some lucid dreamers, despite recognizing that they are dreaming, remain passive observers unable to influence dream content, whereas others can actively shape their actions, emotions, body, dream characters, environment, or broader dream narrative. This distinction between lucid awareness and the ability to intentionally influence dream content is central for both theoretical and applied LD research (Bonamino & Peters, 2025).

Lucid dream control (LDC) can be defined as the capacity to influence dream content during a LD. LDC highly relevant for LD applications. In nightmare treatment, lucidity may already provide therapeutic benefits by allowing dreamers to recognize the dream state as not real, but the ability to alter threatening situations, confront dream figures, regulate fear, or intentionally awaken may further enhance its clinical effectiveness (Holzinger et al., 2020; Spoormaker et al., 2003; Tzioridou et al., 2025). Similarly, motor practice and skill rehearsal during lucid dreaming require initiating, maintaining, and executing specific dream actions in often very specific dream settings (Bonamino et al., 2023; Peters et al., 2023). Experimental paradigms using real-time communication from dreams likewise depend on sustained lucidity, goal-directed behavior, and the ability to perform pre-arranged actions such as eye signals or motor responses (Konkoly et al., 2020; Konkoly et al., 2026; Peters et al., 2025). Thus, dream control is not merely an additional feature of LD but a key factor in determining whether lucidity translates into experimental, clinical, or practical outcomes.

Emerging empirical and phenomenological evidence indicates that LDC is not a unitary construct, but rather multi-dimensional (Bonamino & Peters, 2025). LDC may involve control over cognition (e.g., maintaining lucidity, attention, intentions, emotional stability, deliberate awakening), the dream body and movement, dream characters, and environment, and broader dream physics (e.g., gravity, time, spatial continuity). Significant individual variability in dream control exists, where dreamers may demonstrate strong abilities in some domains while experiencing limited control over others. For example, an individual may successfully stabilize lucidity and comfortably coordinate dream body movements, yet struggle to change the environment, summon characters, or alter dream physics. Different forms of control may involve partially distinct processes, including metacognitive monitoring, attention, emotion regulation, sensorimotor imagery, and the sense of agency. In this context, agency is particularly relevant, as it refers to the deliberate initiation of intentions and actions within the dream state rather than their successful outcome alone (Haggard, 2017; Rosen, 2018; Windt, 2010). LDC may therefore involve both agentic, intentional, goal-directed actions, or more implicit forms of control,

whereby dream events change in line with expectations or beliefs without explicit intention (Bonamino & Peters, 2025).

LDC remains insufficiently operationalized. Existing instruments assess dream control only partially, often as a single component of broader lucidity or LD skill. For example, while the Lucidity and Consciousness in Dreams (LuCiD) scale (Voss et al., 2013) includes a dream control factor, this contains a limited number of items which do not comprehensively differentiate between LDC domains. Similarly, the Lucid Skill Questionnaire (LUSK) (Schredl et al., 2018), Dream Lucidity Questionnaire (DLQ) (Stumbrys et al., 2014), and Frequency and Intensity Lucid Dream (FILD) questionnaire (Aviram & Soffer-Dudek, 2018) include control-related items, but these are embedded within broader assessments of LD frequency, intensity, skill, or awareness. As a result, current measures provide limited information about which specific forms of control are present, how they differ across individuals, and whether control reflects deliberate action or more general changes in dream content. Existing LDC scales and the constructs they measure are summarized in Appendix A.

The lack of specificity in LDC assessment limits both research and application. LDC can vary between individuals, between LDs, and even within a single lucid episode (Mallett et al., 2021). Broad ratings of “dream control” may therefore obscure meaningful differences between control over specific forms or domains. A dreamer who can reliably remain lucid and regulate fear may exhibit a very different control profile from someone who can manipulate the dream environment but struggles to maintain lucidity. Without a standardized and domain-specific instrument, such differences are difficult to quantify and compare.

The present study introduces and validates the Lucid Dream Control Questionnaire (LDCQ), an instrument designed to comprehensively assess trait differences in agentic LDC. The LDCQ is grounded in a multidimensional framework of volitional dream control and assesses seven theoretically defined domains: Own Cognition, Own Emotion, Own Body, Own Movement, Characters, Environment, and Physics. The study aims to evaluate the internal consistency, item performance, convergent validity, and exploratory construct validity of the LDCQ, and to refine the instrument based on quantitative psychometric indicators and qualitative participant feedback. By providing both overall and domain-specific indices of dream control, the LDCQ aims to support more precise assessment of LDC for research and applied contexts. The study has four main objectives:

1. Development and internal validation of the LDCQ, including item analysis and reliability assessment.
2. Validity testing by comparing the LDCQ with control-related items of the LUSK.
3. Qualitative refinement based on open-ended qualitative participant feedback.

4. Exploratory construct validity testing by examining correlations between LDCQ scores and independent dream and sleep variables (LD frequency, dream recall, nightmare frequency, and subjective sleep quality).

2. Methods

2.1 Participants

Participants were recruited primarily from online lucid dreaming communities, including *Reddit* (r/LucidDreaming, r/Dreams), *Dreamviews*, *LD4All*, relevant Facebook and Discord groups, and word of mouth. The inclusion criteria required participants to be at least 18 years old, fluent in English, and to have experienced at least one LD. All participants provided electronic informed consent prior to participation. Participation was voluntary and no compensation was provided.

2.2 Procedure

Recruitment occurred via public online posts inviting lucid dreamers to contribute to the development of standardized dream-control measures. Interested participants accessed a secure online REDCap survey link and provided electronic consent. Following completion of demographic questions, participants self-reported several dream- and sleep-related behaviors, including dream recall (Schredl, 2004), LD and nightmare frequency (Stumbrys et al., 2013), and sleep quality (during the past month) (1 = very poor, 2 = poor, 3 = fair, 4 = good, 5 = very good). They then completed the LUSK (Schredl et al., 2018) and LDCQ. The LDCQ included an optional open-ended item that encouraged participants to provide qualitative feedback on the questionnaire, including wording, conceptual overlap, and interpretability. All other items were mandatory. Participants were assigned a random ID to ensure confidentiality and anonymity. Data were collected between 15 May 2025 and 15 May 2026 and stored on University of Bern secure servers, accessible only to authorized researchers. The study was approved by the Ethics Committee of the University of Bern and conducted in accordance with the Declaration of Helsinki.

2.3 Questionnaire Development and Scoring

The LDCQ was developed to assess perceived agentic LDC abilities across all recalled LDs. Item development was theory-driven and based on current conceptual frameworks of volitional LDC (Bonamino & Peters, 2025). The original questionnaire consisted of 38 items organized into seven theoretically defined domains representing distinct forms of intentional LDC. These domains were selected to reflect both internally directed forms of control (control of the dream self) and externally directed forms of control (control over the broader dream world) (Figure 1):

- Own Cognition Control: maintaining lucidity, focus, and goal-directed intentions.

- Physics: manipulating physical properties of the dream world, including gravity, time, and spatial properties or continuity
- Own Body: transforming or altering one’s dream body or sensory experience.
- Own Movement: executing, coordinating, or sustaining one’s bodily actions and movements.
- Characters: summoning, modifying, interacting with, or influencing dream characters.
- Environment: creating or altering locations, surroundings, weather conditions, broader environmental features, or specific objects.
- Own Emotion: regulating one’s emotional responses and affect and maintaining composure during emotionally intense dream experiences.

Items were phrased as questions (“How often can you deliberately and successfully...”) and rated on a five-point Likert scale (0 = Never, 1 = Rarely, 2 = Sometimes, 3 = Often, 4 = Always). Participants were instructed to indicate how frequently they successfully achieved each type of control when intentionally attempting to do so. An additional response option (“N.A.”) was provided for LDC behaviors that participants had never attempted or considered not applicable to their dream experience. Each dream control domain score was calculated as proportional scores ranging from 0 to 1, with higher scores indicating greater perceived control. Scores were computed as the mean of completed items divided by the maximum possible score (4 for each item), excluding N.A. responses.

2.4 Data Analysis

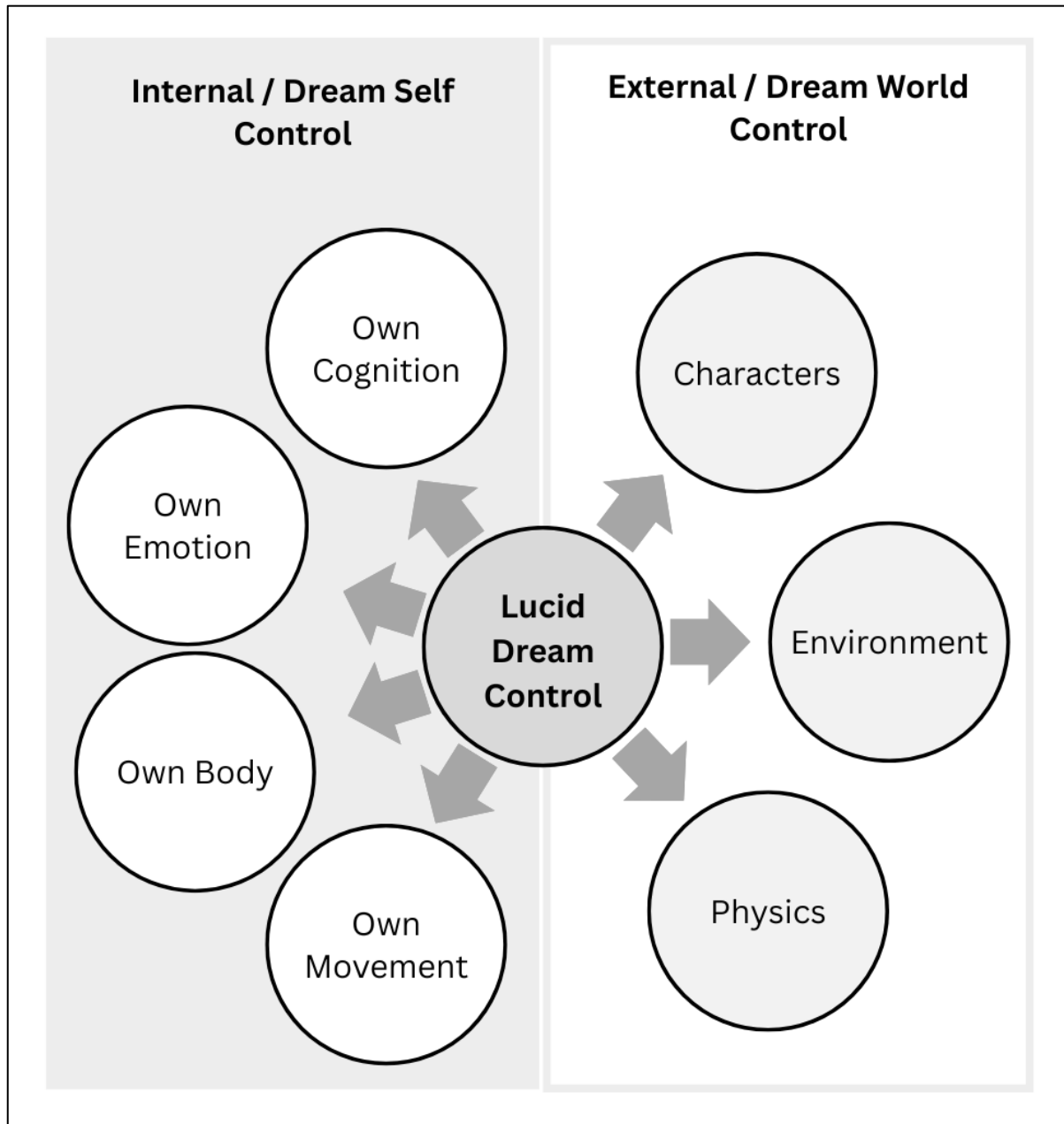
All inferential statistics were conducted on Jamovi (The Jamovi Project, 2025) with significance set at $p < .05$. Results are reported in mean $\pm$ standard deviation unless otherwise stated.

2.4.1 Item-level Evaluation and Internal Consistency

For each item, mean, SD, skewness, and kurtosis were calculated. Items showing minimal variance ($SD < 0.50$), strong skewness ($>|1|$), or kurtosis ($>|2|$) were flagged as potentially uninformative (Kline, 2011). Corrected item–total correlations were calculated to evaluate item contribution to the overall construct. Internal consistency was assessed using Cronbach’s alpha, with $\alpha \geq .70$ considered acceptable (Nunnally & Bernstein, 19954). Inter-item Pearson’s correlations were additionally examined to identify potential redundancy or weak construct alignment.

Figure 1

Theoretical Model of Lucid Dream Control and Domains



Note. The Lucid Dream Control Questionnaire (LDCQ) was developed to assess seven theoretically derived domains of volitional lucid dream control. Domains are conceptually grouped into internally directed forms of control of the dream self (Own Cognition, Own Emotion, Own Body, Own Movement) and externally directed forms of control over the dream world (Characters, Environment, Physics).

2.4.2 Convergent Validity

Convergent validity was assessed by comparing LDCQ domain scores with corresponding LDC domains derived from the LUSK (Schredl et al., 2018). Specifically, LUSK items explicitly assessing dream control (items 4–10; see Table 1) were grouped into a LUSK-Control composite score. Pearson's correlations were calculated between an overall LDCQ score (computed using the same procedure as the domain scores but across all LDCQ items, excluding those marked as N.A.) and domain-specific scores and the LUSK-Control composite. For both the LDCQ and LUSK-Control measures, scores were computed as the mean of completed items.

Table 1

LUSK-Control Composite Items and Corresponding LDCQ Domains

LUSK-Control Composite: "In how many of your lucid dreams..."	Corresponding LDCQ domain
4. ...did you think about different options of what you can do in a dream?	Own Cognition
5. ...were you able to keep your awareness for a satisfying period of time?	Own Cognition
6. ...did you decide deliberately to observe the dream as a dream?	Own Cognition
7. ...were you able to deliberately shape your environment (e.g., change landscapes/surroundings, let persons/characters appear or disappear)?	Environment
8. ...did you have full control of your dream body (movements, actions)?	Own Movement
9. ...did you choose deliberately a specific action?	General
10 ...were you able to perform specific actions (e.g., flying, floating, talking with dream characters, perform magic, engaging in sex)?	General

Note. LUSK: Lucid Dreaming Skills Questionnaire. LDCQ: Lucid Dream Control Questionnaire. The LUSK-Control composite consisted of LUSK items explicitly measuring dream control abilities (LUSK items 6-10).

2.4.3 Construct Validity

To provide preliminary evidence of construct validity beyond psychometric indicators, Pearson's correlations were calculated to explore potential associations between LDCQ scores and dream- and sleep-related variables (dream recall, nightmare frequency, LD frequency, subjective sleep quality).

2.4.4 Qualitative Evaluation

Participant feedback was analyzed using inductive content analysis. Comments were grouped into recurring categories related to item clarity, interpretability, conceptual overlap, and survey usability. Qualitative findings were considered alongside the computed statistics and informed post-validation refinement of the final LDCQ version.

3. Results

3.1 Sample Characteristics

After exclusion of incomplete responses, the final sample consisted of 114 participants (age: 36.4 ± 12.1 years, range: 19–79 years; 53 female). Dream recall frequency was high overall, with 72.8% of participants recalling dreams several times per week or more. LD frequency was similarly high, with 37.7% reporting LDs at least once per week. Nightmare frequency showed greater variability, with most participants reporting nightmares either less than once per year (24.6%) or 2–4 times per year (21.9%). Subjective sleep quality was mostly rated as fair (40.4%) or poor (31.6%). The full distributions for the dream- and sleep-related variables are provided in Appendix B.

3.2 Psychometric Evaluation of the LDCQ

3.2.1 Item Performance

Descriptive statistics for all LDCQ items are presented in Table 2. Item means ranged from 0.60 (Alter sense of smell) to 2.46 (Maintain lucidity without waking), indicating substantial variability in perceived LDC abilities across domains. Standard deviations ranged between 0.98 and 1.45. Overall, item 1 “Maintain lucidity without waking” had the highest mean score (2.46 ± 1.04), followed by item 34 “Remaining calm” (2.41 ± 1.17) and Item 21 “Flight control” (2.40 ± 1.34). Item 17 “Alter sense of smell” had the lowest mean score (0.60 ± 0.98).

The frequency of “N.A.” responses varied considerably, ranging from 0 (Maintain lucidity without waking) to 27 (Alter sense of smell), reflecting differences in the extent to which participants engaged with or attempted specific aspects of dream control. Overall, most items demonstrated acceptable distributional properties. Skewness ranged from -0.47 to 1.79 , and kurtosis from -1.35 to 2.74 . Item 8 slightly exceeded the recommended skewness threshold (skewness = 1.09). Item 17 showed significant skewness (skewness = 1.79) and was therefore examined further during questionnaire refinement.

Table 2*LDCQ Item-level Descriptive Statistics and Distributional Properties*

LDCQ Domain	Item No.	Item Label	M	SD	NA Freq	Skew	Kurt	<i>r</i>
Own Cognition	1	Maintain lucidity	2.46	1.04	0	-0.39	-0.27	0.68
	2	Remember intention	1.94	1.23	2	-0.09	-0.99	0.65
	3	Prolong dream stability	1.77	1.18	3	-0.09	-1.06	0.74
	4	Stay focused on goal	2.21	1.12	1	-0.47	-0.43	0.72
	5	Prevent loss of lucidity	1.92	1.16	5	-0.09	-0.89	0.74
	6	Control without distraction	2.08	1.18	3	-0.30	-0.82	0.77
	7	Overcome distractions	1.98	1.15	6	-0.15	-0.75	0.76
Physics	8	Alter passage of time	0.97	1.20	19	1.09	0.20	0.72
	9	Modify gravity	2.26	1.37	4	-0.40	-1.04	0.66
	10	Third-person perspective	1.33	1.29	15	0.48	-1.08	0.53
	11	Pass through objects	1.74	1.43	8	0.20	-1.28	0.73
	12	Change object properties	1.33	1.28	15	0.61	-0.66	0.73
Own Body	13	Modify dream body	1.60	1.31	9	0.25	-1.09	0.70
	14	Shapeshift / transform	1.12	1.26	17	0.85	-0.30	0.69
	15	Alter visual perception	0.94	1.05	19	0.75	-0.51	0.61
	16	Alter auditory perception	1.00	1.25	24	1.00	-0.13	0.62
	17	Alter sense of smell	0.60	0.98	27	1.79	2.74	0.56
Own Movement	18	Fine motor actions	1.92	1.33	11	0.12	-1.15	0.77
	19	Athletic movements	2.28	1.35	4	-0.32	-0.96	0.65
	20	Balance control	2.28	1.38	11	-0.36	-1.07	0.61
	21	Flight control	2.40	1.34	1	-0.40	-1.04	0.65
	22	Floating	1.91	1.45	8	0.11	-1.35	0.68
	23	Coordinated actions with others	2.38	1.26	13	-0.35	-0.76	0.73
Characters	24	Summon or remove characters	1.68	1.23	5	0.21	-0.91	0.77
	25	Alter character appearance	1.30	1.17	13	0.50	-0.70	0.68
	26	Alter character behavior	1.60	1.15	7	0.12	-0.81	0.72
	27	Casual conversation	2.35	1.35	1	-0.30	-1.06	0.64
	28	Deep conversation	1.90	1.35	2	0.18	-1.10	0.65
Environment	29	Travel to desired location	1.83	1.24	2	0.07	-0.99	0.73
	30	Travel to random location	1.87	1.33	4	0.00	-1.12	0.76
	31	Alter weather	1.07	1.28	16	0.95	-0.22	0.80
	32	Create or remove objects	1.46	1.28	9	0.55	-0.71	0.73
	33	Control lighting	0.97	1.16	17	0.92	-0.21	0.67
Own Emotion	34	Remain calm	2.41	1.17	2	-0.31	-0.89	0.57
	35	Remove fearful element	1.89	1.24	9	0.04	-0.90	0.65
	36	Change emotional state	1.80	1.27	13	0.14	-0.89	0.62
	37	Sustain dream during strong emotions	1.90	1.29	9	-0.05	-1.15	0.74
	38	Maintain control during emotions	1.91	1.26	9	-0.07	-1.09	0.69

Note. Item No. = item number; M = mean; SD = standard deviation; NA Freq = frequency of N.A. responses; *r*: corrected Pearson's item-total correlation. Higher scores indicate greater perceived LDC ability.

3.2.2 LDCQ Item-total and Inter-item Correlations

Corrected item–total correlations indicated that all items contributed positively to the overall construct of LDC. Item–total correlations ranged from .53 to .80 (mean $r = .69$), with no item falling below conventional acceptability thresholds. Inter-item Pearson’s correlations ranged from .10 to .89 and were predominantly moderate to strong, supporting internal coherence across domains. The strongest association was observed between items 37 and 38 ($r = .89$), suggesting conceptual redundancy, and thereby prompted further item revision. Overall, the correlation patterns supported LDCQ’s multidimensional, yet internally coherent, structure.

3.2.3 LDCQ Internal Consistency

Cronbach’s alpha coefficients indicated satisfactory to excellent internal consistency across all dream control domains. Own Cognition ($\alpha = .92$), Own Movement ($\alpha = .92$), Own Emotion ($\alpha = .83$), and Environment ($\alpha = .83$) demonstrated excellent reliability. Characters ($\alpha = .79$), Own Body ($\alpha = .78$), and Physics ($\alpha = .77$) showed acceptable internal consistency. These findings support the internal coherence of the proposed multidimensional structure of the LDCQ.

3.2.4 Domain-level Results

Participants reported the highest average control in Own Movement (2.19 ± 0.06), followed by Own Cognition (2.05 ± 0.06), Own Emotion (1.98 ± 0.04), Characters (1.76 ± 0.09), Physics (1.53 ± 0.08), and Environment (1.44 ± 0.06). The lowest reported control was Own Body (1.05 ± 0.13).

3.2.5 Convergent and Construct Validity

The total LDCQ score showed a strong positive association with the LUSK-Control composite ($r = .73$, $p < .001$), supporting convergent validity. All LDCQ domains were significantly associated with LUSK-Control, with correlations ranging from moderate to strong. The strongest associations were observed for Own Cognition ($r = .74$, $p < .001$), Own Emotion ($r = .69$, $p < .001$), Own Movement ($r = .67$, $p < .001$), and Environment ($r = .63$, $p < .001$), whereas Physics ($r = .59$, $p < .001$), Characters ($r = 0.56$, $p < .001$), and Own Body ($r = .49$, $p < .001$) showed comparatively weaker relationships. These findings indicate that the LDCQ captures constructs that overlap with existing measures of lucid dreaming skill while also differentiating among specific control domains.

Exploratory correlations were conducted to examine relationships between LDCQ scores and related dream and sleep variables. Overall LDCQ scores were positively associated with LD frequency ($r = .63$, $p < .001$) and dream recall frequency ($r = .39$, $p < .001$). No significant associations were observed for nightmare frequency ($r = .03$, $p = .73$) or subjective sleep quality ($r = .12$, $p = .21$). These findings provide preliminary support for LDCQ’s construct validity and suggest that greater perceived dream

control is primarily associated with higher LD frequency and dream recall but not with nightmare frequency or overall subjective sleep quality.

3.3 Post-Analysis Revision

3.3.1 Participant Feedback and Item Revision

Open-ended feedback from 29 participants was analyzed. Responses were clustered into six categories: clarity, interpretation of the N.A. option, personal experiences, REDCap usability, positive feedback, and numerical response issues. Most comments consisted of personal reflections on LD experiences (n = 15), including perceived limits of dream control, changes in LD frequency, and individual induction styles. These comments provided contextual insight into response behavior but did not indicate systematic issues with questionnaire comprehensibility. Comments on clarity (n = 7) led to several targeted refinements. For example, participants, particularly the non-native English speakers, reported difficulty distinguishing between the verbs “maintain” (item 37) and “sustain” (item 38) within the Own Emotion control domain. These items were merged into a single item assessing the ability to maintain lucidity and control during strong emotional LD experiences. Participants additionally identified conceptual overlap between two Environment items assessing transitions to “desired” (item 29) versus “random” (item 30) locations. These items were merged into one broader location-transition item. Additionally, wording was simplified to improve overall questionnaire readability and clarity. For example, a short clarification was added to the Dream Characters domain description, noting that “dream characters” include humans, animals, or other entities. The LDCQ introduction and instructions were also revised for conciseness and clarity, particularly around the use of N.A., where participants are now instructed to mark an item as N.A. if that particular dream control behavior was never attempted. Participants qualitative feedback and resulting changes made to the LDCQ are detailed in Appendix C and D, respectively.

3.3.2 Quantitative Patterns

Quantitative response patterns further informed questionnaire refinement. Items with elevated N.A. response rates and weaker psychometric characteristics were reviewed for conceptual fit and interpretability. For example, item 17, originally assessing alteration of the sense of smell, showed relatively poor distributional properties (mean = 0.60, skewness = 1.79) and high N.A. response rates (N.A. = 27). To improve applicability while retaining the underlying construct, the item was restructured into a broader sensory modulation item encompassing the deliberate alteration of smell, taste, or touch. Overall, conceptually relevant content was retained to preserve coverage across domains. Further, while the original seven-domain structure was retained to reflect the theoretical distinction between different forms of LDC, the order of domain presentation was revised. Domains primarily involving control of the dream-self, or internally directed forms of LDC (Own Cognition,

Own Emotion, Own Body, Own Movement) were represented before domains involving control of external content, or externally directed forms LDC (Characters, Environment, Physics). This reorganization improves conceptual progression and interpretation.

3.4 Final Scoring Framework: Control, Engagement, and Profiling

Response patterns indicated that a single overall LDCQ score did not fully capture differences in how participants engaged with different aspects of LDC. Participants selected the N.A. because they had never attempted a specific form of LDC, not necessarily because they were incapable of performing it. Treating N.A. responses as indicators of low control could therefore underestimate an individual's LDC abilities. To address this issue, N.A. responses were retained but excluded from control-score calculations. LDCQ scores thus reflect perceived control over the forms of LDC that participants have attempted, rather than penalizing them for forms they never tried. To complement these scores, a dream engagement score was introduced for each domain to quantify the breadth of LDC behaviors participants had attempted, independent of their perceived control ability. Engagement scores were calculated as the proportion of non-N.A. responses relative to the total number of items within each domain and range from 0 (all items marked as N.A.) to 1 (no items marked as N.A.). Higher engagement scores, therefore, reflect greater reported engagement with that LDC domain.

The scoring framework was further extended to generate a multi-dimensional LDC profile rather than relying exclusively on a single global score. The LDCQ yields seven independent domain scores – Own Cognition, Own Body, Own Movement, Own Emotion, Characters, Environment, and Physics – alongside corresponding dream engagement scores for each domain. Mean dream control and engagement scores for the present sample are presented in Table 3 and can be visualized using a multidimensional spider diagram (Figure 2). Control varied across domains, with the highest mean score observed for Own Movement, followed by Own Cognition and Own Emotion, Characters, Physics, Environment, and Own Body. Engagement was consistently high across domains, indicating that participants had attempted a broad range of dream control behaviors despite considerable variability in their perceived success. This profile-based representation emerged from the observation that participants exhibited heterogeneous patterns of LDC and engagement despite similar global dream control scores. By displaying both dream control and dream engagement, the profile distinguishes between the breadth of dream control behaviors participants attempt and their perceived ability to successfully perform them. Together, these complementary metrics provide a more nuanced characterization of LDC than a single aggregate score.

Table 3

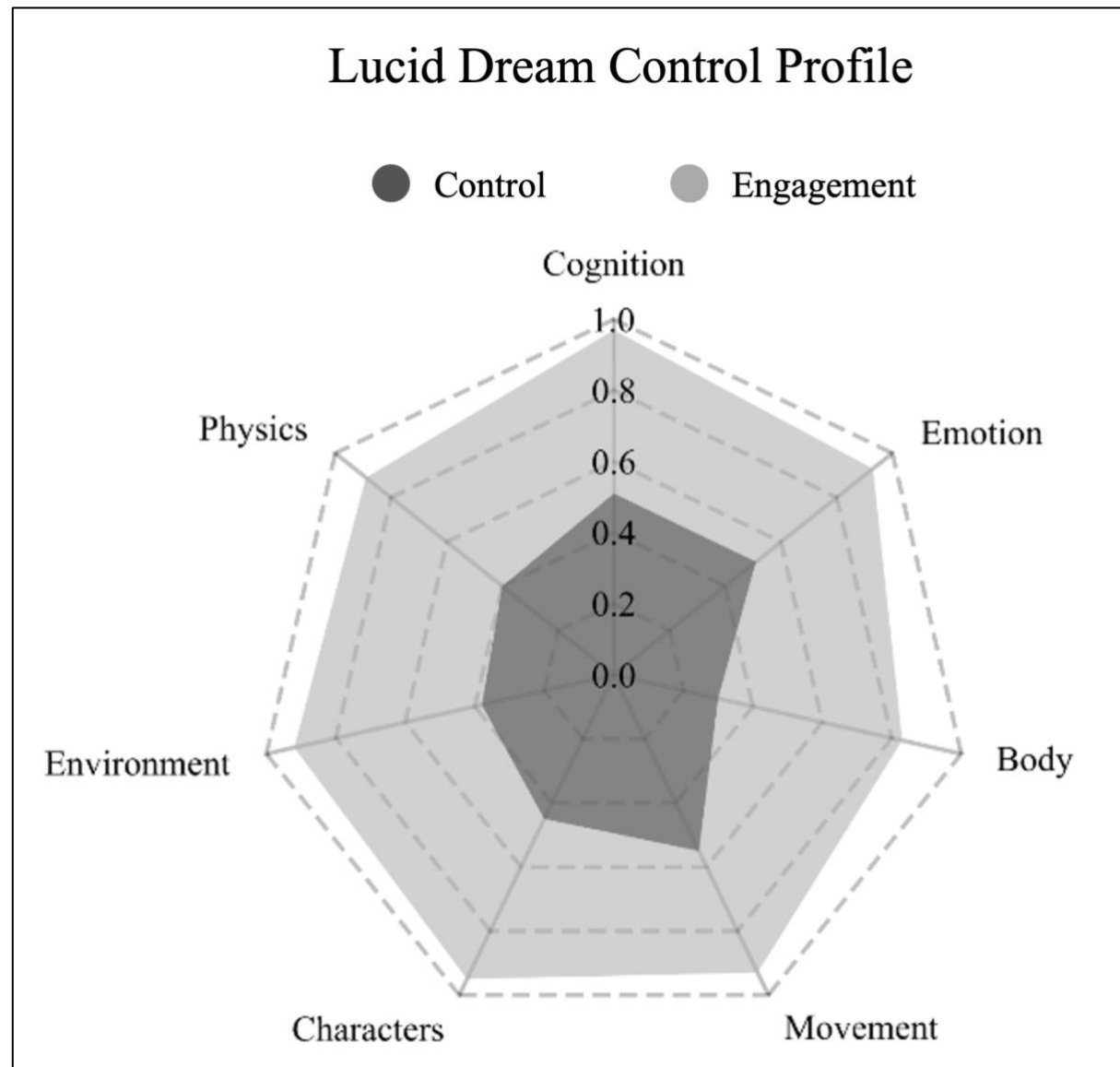
Mean Dream Control and Dream Engagement Scores Across LDCQ Domains

Dream Control Domain	Dream Control (M ± SD)	Dream Engagement (M ± SD)
Own Cognition	0.51 ± 0.24	0.97 ± 0.09
Own Emotion	0.51 ± 0.25	0.93 ± 0.17
Own Body	0.30 ± 0.26	0.83 ± 0.31
Own Movement	0.55 ± 0.26	0.93 ± 0.16
Characters	0.45 ± 0.25	0.95 ± 0.15
Environment	0.38 ± 0.27	0.92 ± 0.19
Physics	0.40 ± 0.26	0.89 ± 0.22

Note. M = mean; SD = standard deviation.

Figure 2

Lucid Dream Control Profile Based on Mean LDCQ Domain Control and Engagement Scores



Note. Figure 2 shows the LDC profile of this study's sample. The dark polygon represents domain-specific dream control scores on Lucid Dream Control Questionnaire (LDCQ), whereas the light polygon represents domain-specific dream engagement scores. Control scores reflect participants' perceived ability to successfully exert dream control over dream elements they have attempted to influence within each domain. Engagement scores indicate the proportion of non-N.A. responses relative to the total number of items within each domain, thereby reflecting the extent to which participants attempted the corresponding forms of dream control. Scores range from 0 to 1, with higher values indicating greater control or engagement.

To facilitate interpretation and application of the LDCQ, an accompanying online scoring tool was developed that automatically calculates dream control and dream engagement scores and generates an individualized LDC profile (Bonamino & Bonamino, 2026). The final LDCQ items are summarized in Table 4, while the Supplementary Material contains the complete participant-facing version, including instructions scoring guide, and link to the online assessment tool.

Table 4

Final Version of the Lucid Dream Control Questionnaire (LDCQ)

Domain	Item No.	Item "How often can you deliberately and successfully..."
Own Cognition	1	...maintain lucidity without waking up?
	2	...remember a pre-set intention while lucid?
	3	...prolong the lucid dream when it starts to fade or feel unstable?
	4	...remain focused on a chosen goal or task without distraction?
	5	...prevent losing lucidity or waking before completing an intended action or intention?
	6	...experience dream control without major distractions or instability (e.g., struggling to exert control, being distracted by the dream or dream characters)
	7	...regain focus or control when distractions or instability occur?
Own Emotion	8	...remain calm and collected during overwhelming, stressful, or highly exciting dream situations?
	9	... remove an element that induce fear, stress, or unwanted emotions?
	10	...change your emotional state (e.g., induce joy or relaxation)?
	11	...maintain lucidity and control when experiencing strong emotions (e.g., excitement, fear, frustration)?
Own Body	12	...modify your dream body (e.g., size, shape, abilities)?
	13	...shapeshift or transform into another being (e.g., animal, plant, person, fictional character) or inanimate object?
	14	...disable or alter your visual perception (e.g., black and white vision, looking through binoculars)?

Domain	Item No.	Item “How often can you deliberately and successfully...”
	15	...disable or alter auditory perception (e.g., make sounds feel louder or softer)?
	16	...disable or alter other sensory perception (e.g., intensity or quality of smell, taste, touch)?
Own Movement	17	...perform fine motor actions with precision such as writing, playing an instrument, or using tools?
	18	...perform athletic movements with ease such as sprinting, climbing, or gymnastics?
	19	...maintain balance while performing actions like jumping, standing on one foot, or walking on narrow surfaces?
	20	...take off, fly, and control your flight direction and altitude (e.g., turning mid-air, adjusting speed, stopping)?
	21	...float (not fly)?
	22	...perform coordinated motor actions with other dream characters (e.g., dancing, fighting, playing sports together)?
Characters	23	...summon or remove a specific dream character?
	24	...alter the appearance of a dream character (e.g., change their clothes, height, hair, other physical features)?
	25	...alter the behavior of a dream character (e.g., ordering them to complete an action or behave in a specific way)?
	26	...have a casual conversation with a dream character (e.g., exchanging greetings or small talk)?
	27	...have a deep conversation with a dream character (e.g., seeking life advice, new knowledge, solving a problem)?
Environment	28	...travel or transition to another location by actively altering your surroundings (e.g., teleporting, using a portal, turning around, turning corners, wishing them into existence)?
	29	...alter the weather or climate (e.g., making it rain or snow)?
	30	...create or remove objects (e.g., trees, buildings, specific elements of your surroundings)?
	31	...control the lighting (e.g., make the dream space brighter or darker, changing the hue/color)?
Physics	32	...alter the passage of time (e.g., speed up, slow down, pause, rewind time)?
	33	...modify gravity (e.g., making objects, yourself, or others float or fly, reducing or increasing gravity)?
	34	...switch viewpoint (e.g., from first-person to observer perspective)?
	35	...pass through solid objects or walls?
	36	...change fundamental properties of objects or surroundings (e.g., density, solidity)?

333 *Note.* The final Lucid Dream Control Questionnaire (LDCQ) comprises 36 items across seven domains
334 following post-validation refinement. Domains are organized in internally directed aspects of LDC
335 (Own Cognition, Own Emotion, Own Body, Own Movement) and externally directed aspects of LDC
336 (Characters, Environment, Physics). During refinement, the original items 29-30 and items 37-38 were

merged, reducing the total item count from 38 to 36. The LDCQ yields domain-specific control scores, domain-specific engagement scores, and an overall dream-control profile (Figure 2).

4. Discussion

4.1 Summary of Main Findings

The present study introduced and validated the LDCQ, a multidimensional instrument designed to assess perceived dream control abilities across LDs. Overall, the questionnaire demonstrated good psychometric performance, including strong internal consistency across domains, meaningful item–total associations, and strong convergent validity with an established measure of lucid dreaming skills. Together, these findings provide initial support for the LDCQ as a standardized instrument for assessing individual differences in LDC. Across domains, participants reported the highest levels of dream control for movement- and cognition-related abilities, followed by emotion, dream character, environment, and physics. Dream body control received the lowest average scores. This pattern broadly aligns with previous evidence suggesting that dreamers are more readily able to maintain awareness, move their dream body, and interact with dream characters and environment than perform abstract or extensive transformations of dream content (Kahan, 1994; Peters et al., 2025; Stumbrys et al., 2014). Notably, the comparatively lower body scores may partly reflect the inclusion of more uncommon sensory-control items (altering the sensory perception of smell, taste, and touch) introduced in the present instrument.

4.2 Convergent and Construct Validity

The strong association between the LDCQ and the LUSK-Control composite supports the convergent validity of the LDCQ and suggests that it captures core aspects of LDC identified by other established measures. At the same time, the domain-specific structure extends beyond existing instruments by distinguishing between multiple dimensions of LDC rather than treating it as a single, broader construct. Exploratory associations with dream and sleep variables are broadly consistent with theoretical expectations and provide additional evidence for the construct validity of the LDCQ. Higher LDCQ scores were associated with greater LD and dream recall frequency. Dream recall is a well-known predictor of LD frequency (Kelly et al., 2022; Schredl & Erlacher, 2004, 2011; Schredl et al., 2022) and higher LD frequencies have been associated with greater dream control (Sammer et al., 2024; Schredl et al., 2018; Stumbrys & Erlacher, 2017). This is expected as individuals who experience LDs more frequently have greater opportunities to develop and refine dream control abilities through repeated practice. In contrast, associations with nightmare frequency and subjective sleep quality were small and non-significant, indicating that perceived dream control may be independent of both nightmare occurrence and general sleep quality. LDC may be more closely related to outcomes such as nightmare management, distress reduction, or emotional regulation, than to the occurrence of nightmares

themselves, variables that were not assessed in the present study. Consistent with this, evidence from lucid dreaming therapy suggests that the ability to control dream content may enhance therapeutic outcomes beyond attaining lucidity alone (Holzinger et al., 2020; Spoormaker et al., 2003; Tzioridou et al., 2025). Further, low-control LDs are associated with more negative outcomes compared to those with greater LDC (Mallet et al., 2022). These findings should however be interpreted with caution, given the relatively small sample size of the present study and that association analyses were conducted using the pre-refinement version of the questionnaire.

4.3 Theoretical Implications

The present findings support the view that LDC is not a unitary construct but rather comprises multiple related, yet distinct dimensions of volitional control of LDs (Bonamino & Peters, 2025; Mallett et al., 2021). Different domains of LDC may rely on partially distinct cognitive and neural processes. For example, cognitive and emotion control may rely on metacognitive awareness, attentional regulation, and executive stability (Double, 2026; Filevich et al., 2015; Mansueto et al., 2024), whereas dream body and movement control may rely more heavily on sensorimotor simulation networks active during Rapid Eye Movement (REM) sleep (Dresler et al., 2011; Noreika et al., 2020; Speth & Speth, 2016). Importantly, the LDCQ extends beyond other existing questionnaires by operationalizing LDC as a multidimensional construct. Rather than treating LDC as a single, broad ability, the LDCQ distinguishes between different domains in which control can be expressed. This multidimensional approach may prove valuable for future research investigating whether different forms of dream control show unique relationships with neural activity, personality characteristics, LD interventions, or other clinical or non-clinical LD applications.

4.4 Practical Applications and Scoring Recommendations

A key advantage of the LDCQ is its flexibility, which allows dream control to be examined at multiple levels of specificity, depending on the intended application. The questionnaire may be used both as a research instrument to investigate individual differences in LDC and as a practical tool for lucid dreamers seeking to better understand their own patterns of dream control abilities.

For researchers, the LDCQ provides several complementary levels of analysis. At the global level, the total dream control score can be computed and used to examine broad relationships between LDC and external variables such as LD frequency, personality characteristics, clinical outcomes, the effects of LD interventions, or other relevant variables of interest. At the domain level, the LDCQ distinguishes between seven core dimensions of dream control (Own Cognition, Own Emotion, Own Body, Own Movement, Characters, Environment, and Physics), allowing researchers to investigate more targeted processes, such as whether different dimensions of LDC show distinct relationships with cognitive, behavioral, neural, or applied outcomes (e.g., emotion control during nightmares, environment control

in creativity studies, movement control in LD motor rehearsal). At the item level, individual item scores may be further examined to study highly specific dream control abilities, such as flight, time manipulation, body transformation, or maintaining lucidity during emotionally intense LD episodes. Item-level analysis may be particularly useful in experimental or intervention studies targeting specific dream behaviors.

The questionnaire can additionally be administered across different temporal scales. When completed with reference to LDs in general, scores provide an estimate of trait-level LDC tendencies. When completed with reference to a specific LD episode, the same item set can be used to assess state-level control and may be combined with qualitative dream reports to capture the characteristics of individual LD experiences. Such flexibility is particularly relevant given that LDC varies not only between individuals, but also across and within LD episodes (Mallett et al., 2021; Peters et al., 2025). A dreamer may, for example, display a broad and highly developed dream control profile in one LD while experiencing a more limited profile in another, or show fluctuations across different phases of the same dream. Although the present study focused primarily on trait-level assessment, future research may benefit from adapting and validating the questionnaire for state-level administration. One approach could involve retaining the existing LDCQ items and scoring framework while revising the wording to reflect its temporal focus. For example, trait-level administration could use the wording, “To what extent are you typically able to deliberately and successfully...”, whereas state-level administration could use, “To what extent were you able to deliberately and successfully...”. Both versions could be rated using the same five-point Likert scale (0 = Not at all, 1 = Slightly, 2 = Moderately, 3 = Quite a bit, 4 = Completely). Such an approach would facilitate direct comparisons between habitual dream control tendencies and control expressed during specific LD episodes, providing a more comprehensive understanding of the dynamic and multidimensional nature of LDC.

Beyond research applications, the LDCQ may also serve as a practical self-assessment tool for lucid dreamers. Domain scores can help individuals identify relative strengths and limitations across different aspects of dream control, potentially informing future LDC practice, training, or exploration. Finally, the LDCQ provides a dream engagement metric that is conceptually distinct from perceived control ability. Whereas control scores reflect how successfully participants report exerting volitional dream control, engagement scores reflect the breadth of LDC behaviors they have attempted. Together, dream control and engagement scores can be visualized as a multidimensional LDC profile, providing both researchers and individual lucid dreamers with a more nuanced characterization of their LD experiences. An accompanying online scoring and visualization tool further facilitates practical application of the LDCQ by automatically generating individualized LDC profiles (Bonamino & Bonamino, 2026).

4.5 Limitations and Future Directions

Several limitations should be acknowledged. First, the relatively small sample size did not permit robust psychometric analyses such as exploratory or confirmatory factor analysis. Validation, therefore, focused on item performance, reliability, and preliminary convergent and construct validity. Second, post-validation refinements were introduced after data collection and were not independently revalidated in a second sample. Nonetheless, the questionnaire demonstrated good psychometric performance across multiple indicators, including strong internal consistency, meaningful item-total associations, and strong convergent validity with an existing measure of lucid dreaming skills. Together, these findings provide encouraging preliminary support for the validity and utility of the LDCQ.

Future research should replicate these findings in larger and more diverse samples, more formally examine the dimensional structure of the LDCQ, and evaluate test–retest reliability. Further work should also examine the validity and utility of state-based adaptations of the LDCQ, as well as the proposed dream control and dream engagement profile framework. The present questionnaire focuses on agentic forms of LDC involving deliberate attempts to influence dream experiences. Future research may benefit from developing complementary measures to assess non-agentic forms of LDC, in which dream content changes in accordance with expectations, beliefs, or intentions without explicit volitional control.

5. Conclusion

The present study introduced the LDCQ, a theoretically grounded and psychometrically supported instrument for assessing LDC. Consistent with emerging theoretical perspectives, the findings support the view that LDC is a multidimensional construct comprising multiple related yet distinct domains of volitional control during LDs. By distinguishing between dream control domains and incorporating both control and engagement metrics, the LDCQ provides a more comprehensive characterization of LDC than existing measures. Beyond offering a standardized tool for assessing individual differences in LDC, the LDCQ provides a foundation for investigating how different dimensions of dream control relate to the mechanisms, applications, and phenomenology of lucid dreaming. As such, the LDCQ represents an important step toward a more nuanced and systematic understanding of LDC across both research and applied contexts.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT (OpenAI, 2025). The tool was used solely to help improve text clarity. After using this service, the authors reviewed and edited as needed and take full responsibility for the content published article.

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